

University of Calcutta
Department of Computer Science & Engineering
Midterm Examination, 2025

Subject: Computer Science & Engineering
Paper Code & Name: CSCL601 & FORMAL LANGUAGE & AUTOMATA THEORY

Time: 1 hour

1.
 - i. What is a Finite State Machine? What are its implications? What are alternative forms?
 - ii. What are equivalent and compatible states? What are their implications?
 - iii. For a given lossless FSM of order 3, how will define the state of the corresponding inverse machine?
 - iv. Justify the statement – 'Equivalence partition of completely specified FSM is unique.'
2.
 - i. Design a single track-single-head Turing machine to compute the multiplication of two integer numbers, represented by unary symbol sets. Clearly state your assumptions.
 - ii. Draw the state transition diagram for deterministic/nondeterministic finite automata accepting the following languages:
 - A. $(01 + 1)^* + (11 + 0)^*$
 - B. $((01 + 001)^* 0^*)^*$
 - iii. Give the regular expressions generating the following languages. In all cases, the alphabet is $\{0, 1\}$.
 - A. $L_1 = \{w \mid w \text{ does not contain } 100 \text{ as a substring}\}$
 - B. $L_2 = \{w \mid w \text{ starts with zero and has odd length or starts with one and has even length}\}$
 - iv. Convert the following regular expression into an equivalent NFA
 $((00)^* (11) + (01))^*$

[4+3+3+2]